

**SAFETY DATA SHEET**

Creation Date 21-January-2009

Revision Date 03-April-2024

Revision Number 2

**1. Identification**

**Product Name** 1-Butanol

**Cat No. :** C10769

**CAS-No** 71-36-3  
**Synonyms** n-Butanol; n-Butyl alcohol, Butan-1-ol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                   |            |
|-------------------------------------------------------------------|------------|
| <b>Flammable liquids</b>                                          | Category 3 |
| <b>Acute oral toxicity</b>                                        | Category 4 |
| <b>Skin Corrosion/Irritation</b>                                  | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>                          | Category 1 |
| <b>Specific target organ toxicity (single exposure)</b>           | Category 3 |
| Target Organs - Respiratory system, Central nervous system (CNS). |            |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Flammable liquid and vapor

Harmful if swallowed  
Causes skin irritation  
Causes serious eye damage  
May cause respiratory irritation  
May cause drowsiness and dizziness

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting/equipment  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Use non-sparking tools  
Take action to prevent static discharges

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
Take off contaminated clothing and wash it before reuse

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component | CAS-No  | Weight % |
|-----------|---------|----------|
| 1-Butanol | 71-36-3 | 99       |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

|                                        |                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water.                                                 |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                        |

## 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | 35 °C / 95 °F                                                                         |
| <b>Method -</b>                         | CC (closed cup)                                                                       |
| <b>Autoignition Temperature</b>         | 340 °C / 644 °F                                                                       |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | 11.2 vol %                                                                            |
| <b>Lower</b>                            | 1.4 vol %                                                                             |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Use personal protective equipment as required. Ensure adequate ventilation. |
| <b>Environmental Precautions</b> | Should not be released into the environment.                                |

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.                                                                                             |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area. Incompatible Materials. Strong oxidizing agents. Reducing Agent. Acid chlorides. copper. Copper alloys. Acid anhydrides. |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component | Alberta                                  | British Columbia               | Ontario TWAEV | Quebec                                                    | ACGIH TLV   | OSHA PEL                                                                                                                    | NIOSH                                                               |
|-----------|------------------------------------------|--------------------------------|---------------|-----------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1-Butanol | TWA: 20 ppm<br>TWA: 60 mg/m <sup>3</sup> | TWA: 15 ppm<br>Ceiling: 30 ppm | TWA: 20 ppm   | Ceiling: 50 ppm<br>Ceiling: 152 mg/m <sup>3</sup><br>Skin | TWA: 20 ppm | Skin (Vacated)<br>Ceiling: 50 ppm (Vacated)<br>Ceiling: 150 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 300 mg/m <sup>3</sup> | IDLH: 1400 ppm<br>Ceiling: 50 ppm<br>Ceiling: 150 mg/m <sup>3</sup> |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments                                         |
|----------------|-------------------|-----------------|--------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.35 mm         | As tested under EN374-3                                |
| Nitrile rubber | > 480 minutes     | 0.38 mm         | Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                           |
|----------------------------------------|---------------------------|
| Physical State                         | Liquid                    |
| Appearance                             | Colorless                 |
| Odor                                   | Alcohol-like              |
| Odor Threshold                         | No information available  |
| pH                                     | No information available  |
| Melting Point/Range                    | -89 °C / -128.2 °F        |
| Boiling Point/Range                    | 117.6 °C / 243.7 °F       |
| Flash Point                            | 35 °C / 95 °F             |
| Method -                               | CC (closed cup)           |
| Evaporation Rate                       | 0.46                      |
| Flammability (solid,gas)               | Not applicable            |
| Flammability or explosive limits       |                           |
| Upper                                  | 11.2 vol %                |
| Lower                                  | 1.4 vol %                 |
| Vapor Pressure                         | 6.7 mbar @ 20 °C          |
| Vapor Density                          | 2.6                       |
| Specific Gravity                       | 0.810                     |
| Solubility                             | Slightly soluble in water |
| Partition coefficient; n-octanol/water | No data available         |
| Autoignition Temperature               | 340 °C / 644 °F           |
| Decomposition Temperature              | No information available  |
| Viscosity                              | 2.95 mPa.s (20 °C)        |
| Molecular Formula                      | C4 H10 O                  |
| Molecular Weight                       | 74.12                     |
| Refractive index                       | 1.390 - 1.400             |

## 10. Stability and reactivity

|                                  |                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                      |
| Stability                        | Stable under normal conditions.                                                                 |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.        |
| Incompatible Materials           | Strong oxidizing agents, Reducing Agent, Acid chlorides, copper, Copper alloys, Acid anhydrides |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                         |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                        |
| Hazardous Reactions              | None under normal processing.                                                                   |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation             |
|-----------|--------------------------|------------------------------|-----------------------------|
| 1-Butanol | LD50 = 700 mg/kg ( Rat ) | LD50 = 3402 mg/kg ( Rabbit ) | LC50 > 8000 ppm ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|               |                                                                |
|---------------|----------------------------------------------------------------|
| Irritation    | Severe eye irritant; Irritating to respiratory system and skin |
| Sensitization | No information available                                       |

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component | CAS-No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|---------|------------|------------|------------|------------|------------|
| 1-Butanol | 71-36-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Central nervous system (CNS)

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not flush into surface water or sanitary sewer system.

| Component | Freshwater Algae                                                                                                                                                                        | Freshwater Fish                                                                                                                                                                                                                                                                                                                | Microtox                                                                                              | Water Flea                                                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-Butanol | EC50: 225 mg/L, 96h<br>(Pseudokirchneriella subcapitata) OECD Guideline 201<br>EC50: > 500 mg/L, 72h<br>(Desmodesmus subspicatus)<br>EC50: > 500 mg/L, 96h<br>(Desmodesmus subspicatus) | LC50: 1376 mg/L, 96h<br>(Pimephales promelas)<br>OECD Guideline 203 :<br>100000 - 500000 µg/L, 96h<br>static (Lepomis macrochirus)<br>LC50: = 1740 mg/L, 96h<br>flow-through (Pimephales promelas)<br>LC50: = 1910000 µg/L, 96h<br>static (Pimephales promelas)<br>LC50: 1730 - 1910 mg/L,<br>96h static (Pimephales promelas) | EC50 = 2041.4 mg/L 5 min<br>EC50 = 2186 mg/L 30 min<br>EC50 = 3980 mg/L 24 h<br>EC50 = 4400 mg/L 17 h | EC50: 1328 mg/L, 48h<br>(Daphnia magna) OECD Guideline 202<br>EC50: 1897 - 2072 mg/L,<br>48h Static (Daphnia magna)<br>EC50: = 1983 mg/L, 48h<br>(Daphnia magna) |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component | log Pow |
|-----------|---------|
| 1-Butanol | 1       |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component           | RCRA - U Series Wastes | RCRA - P Series Wastes |
|---------------------|------------------------|------------------------|
| 1-Butanol - 71-36-3 | U031                   | -                      |

## 14. Transport information

**DOT**

UN-No UN1120  
 Proper Shipping Name BUTANOLS  
 Hazard Class 3  
 Packing Group III

**TDG**

UN-No UN1120  
 Proper Shipping Name BUTANOLS  
 Hazard Class 3  
 Packing Group III

**IATA**

UN-No UN1120  
 Proper Shipping Name BUTANOLS  
 Hazard Class 3  
 Packing Group III

**IMDG/IMO**

UN-No UN1120  
 Proper Shipping Name BUTANOLS  
 Hazard Class 3  
 Packing Group III

## 15. Regulatory information

**International Inventories**

| Component | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 1-Butanol | 71-36-3 | X   | -    | X    | ACTIVE                                        | 200-751-6 | -      | -   |

| Component | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|---------|-------|----------|------|------|------|------|-------|-------|
| 1-Butanol | 71-36-3 | X     | KE-03867 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| 1-Butanol | Part 1, Group A Substance Part 4 Substance           |                                                                            |                                           |

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1-Butanol | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| 1-Butanol | 71-36-3 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 1-Butanol | 71-36-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**